

Changing states

MATTER CHANGES FROM ONE STATE TO ANOTHER ACCORDING TO TEMPERATURE AND PRESSURE.

SEE ALSO

◀ 96–97 Properties of materials

◀ 98–99 States of matter

Convection currents

189 ▶

Every substance has a standard state. This is its state (solid, liquid, or gas) at 25°C (77°F)—just above room temperature. Increasing or decreasing that temperature eventually leads to a change in state.

States and energy

Changes in state are the result of energy being added to or removed from a substance. Taking energy from a gas results in it becoming a liquid and then a solid. Adding energy has the reverse effect. The energy within a substance makes its basic units—atoms or molecules—vibrate (wobble). This vibration, called internal energy, is measured when temperatures are recorded.

SEAgel is a spongelike solid made from seaweed. It is so light that it floats in thin air!

▷ Melting and boiling point

The temperatures at which a solid changes into a liquid or gas are called, respectively, the melting and boiling points. The temperatures are specific to each substance. They are always measured at standard atmospheric pressure. Changes in pressure affect the temperatures at which substances change state.

Sublimation

Sometimes a solid does not melt, but turns straight into a gas in a process called sublimation. Carbon dioxide sublimates almost all the time, while water ice changes straight into vapour if the air is very dry.

Deposition

The opposite process to sublimation, deposition, occurs when a gas turns into a solid without first becoming a liquid. Ice can be deposited from the vapor in air in very cold conditions.

Freezing

In a liquid, the vibration or internal energy is just enough to break a few bonds—in fact, they are constantly breaking and reforming. A liquid freezes into a solid when its atoms or molecules no longer have enough energy to keep breaking bonds.

REAL WORLD

Salting ice

Adding salt to ice lowers the melting point of water by a couple of degrees. Salting roads in winter stops dangerous sheets of ice from forming—although if the conditions are well below 0°C (32°F) the water will still freeze. The salt dissolved in the water gets in the way of the water molecules, making it harder for them to form all the bonds they need to become ice.

